EE2001 Electrical Circuits- I:

Continuation of the topic:
Techniques of Circuit Analysis

11 December 2023

Outline



Continuation of the topic:

10. Thevenin and Norton Equivalents

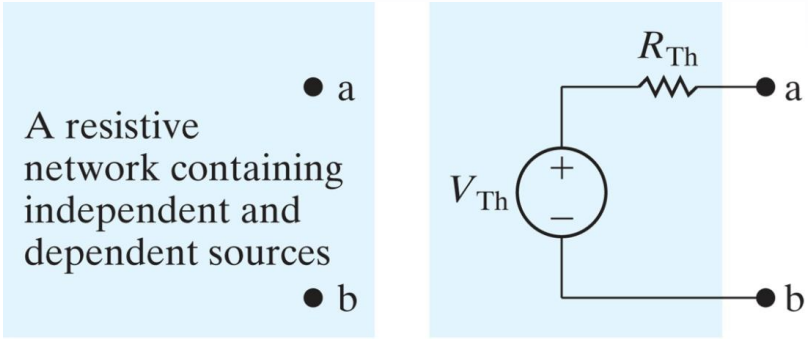
11. Maximum Power Transfer

Thevenin and Norton Equivalents

- The Thevenin and the Norton equivalents are **circuit simplification** techniques that focus on the **terminal behavior** of circuits.
- They can be used in circuits composed of **linear elements**.

The Thevenin Equivalent

- Consider any circuit composed of sources (both dependent and independent) and resistors.
- The **Thevenin equivalent** is equivalent to the original circuit in the sense that if the same load resistor is connected to both networks/circuits **the current and voltage in both are the same**.

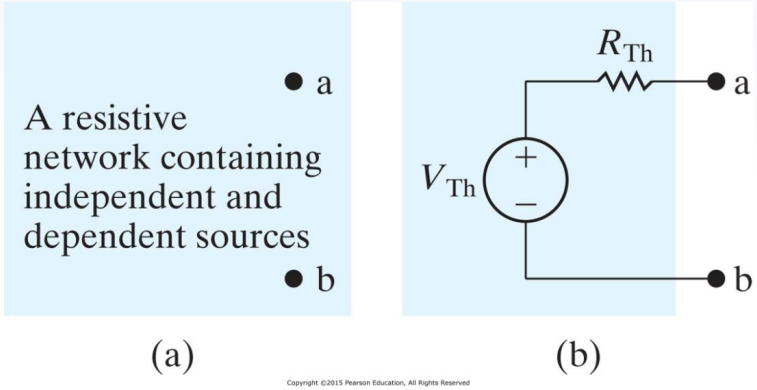


Copyright ©2015 Pearson Education, All Rights Reserved

- In other words, **their terminal current-voltage characteristics are the same**.
- The equivalence holds **for all possible values of the load resistance**.

The Thevenin Equivalent

- I order to determine the **Thevenin equivalent** of a given circuit, we need to determine $\boldsymbol{V_{Th}}$ and $\boldsymbol{R_{Th}}$



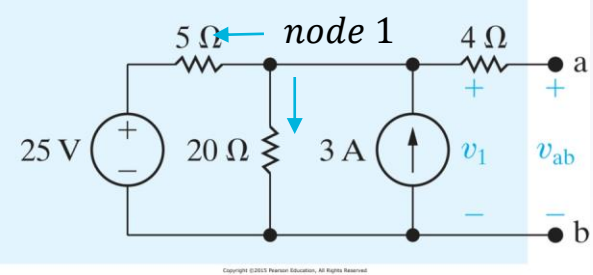
Copyright ©2015 Pearson Education, All Rights Reserved

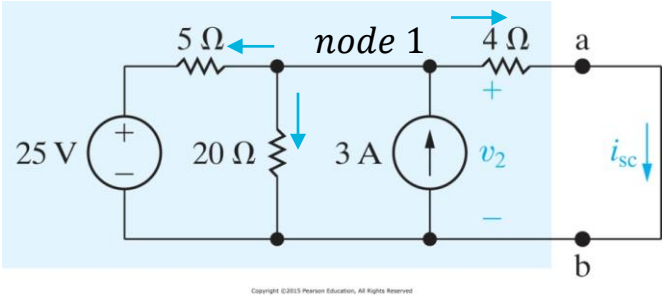
- Considering the **open circuit** and **short circuit conditions**, we can derive:
  - $V_{Th} = V_{Oc}$ =Open circuit voltage
  - $R_{Th} = \frac{V_{Oc}}{I_{Sc}}$ = Open circuit voltage/Short circuit current

The Thevenin Equivalent

10

- Example:





- We first calculate the **open-circuit** voltage of v_{ab}
- When the terminals a nd b are **open**, there is **no current** in the 4 Ω resistor. Therefore the open circuit voltage v_{ab} is **identical** to the voltage across the 3 A current source.
- By solving **a node-voltage equation**, choosing the lower node as the reference node,

$$\frac{v_1-25}{5}+\frac{v_1}{20}-3=0 \qquad v_1=32\,\text{V}=V_{Th}$$

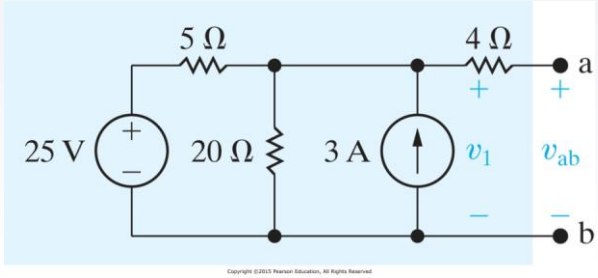
- The next step is to place **a short circuit** across the terminals and calculate the resulting short-circuit current. Short-circuit current **is in the direction** of the open-circuit voltage drop across the terminals a, b.
- Once v_2 is **known**, i_{sc} **can be found** easily. Again using the lower node as the reference node,

$$\frac{v_2-25}{5}+\frac{v_2}{20}-3+\frac{v_2}{4}=0 \qquad v_2=16\,\text{V} \qquad i_{sc}=\frac{16}{4}=4\,A$$

The Thevenin Equivalent

10

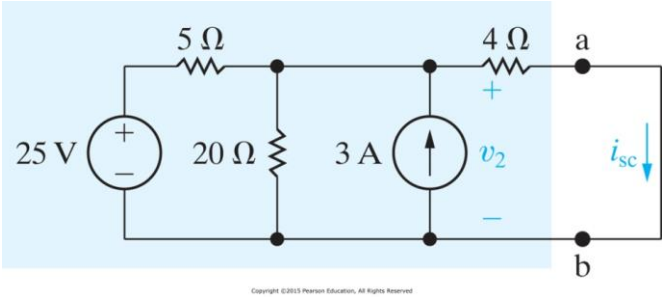
- Example:

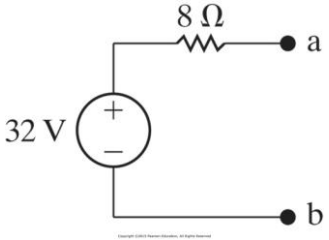


$$V_{Th} = 32\ V$$

$$i_{sc} = 4\ A$$

$$R_{Th} = \frac{V_{Th}}{i_{Sc}} = \frac{32}{4} = 8\ \Omega$$





The Norton Equivalent

10

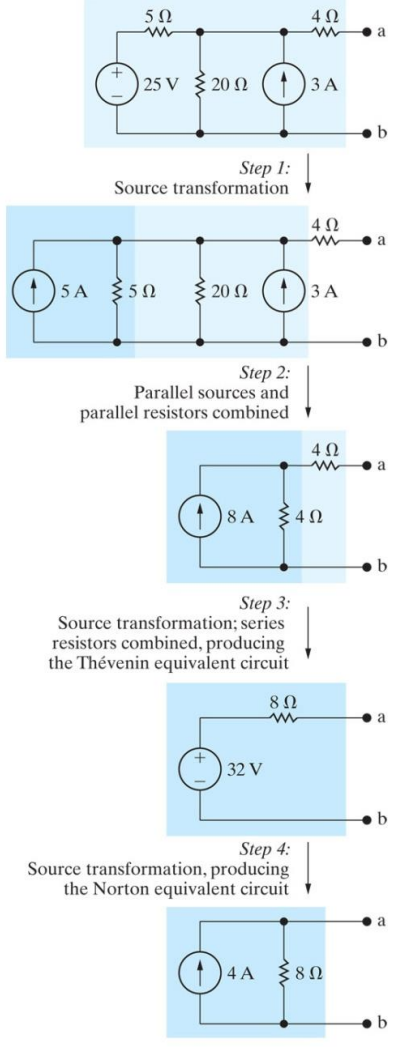
- The Norton equivalent is an **independent current source in parallel with a resistor**.
- In order to determine the Norton equivalent of a given circuit, we need to determine $\boldsymbol{I_N}$ and $\boldsymbol{R_N}$.
- Considering the **open circuit** and **short circuit** conditions, we can derive that,
  - $I_N = I_{Sc}$= Short circuit current
  - $R_N = R_{Th} = \frac{V_{Oc}}{I_{Sc}}$ = Open circuit voltage / Short circuit current
- **Similar** to the Thevenin equivalent, the Norton equivalent is equivalent to the original circuit in the sense that if the same load resistor is connected to both networks/circuits the **current** and **voltage** in both are the same.
- In other words, their **terminal current-voltage characteristics are the same**.
- The equivalence holds for **all possible values of the load resistance**.

The Norton Equivalent

- The Norton equivalent can also easily be **derived from the Thevenin equivalent** by a **source transformation**.
- It is also possible to **utilize source transformations** for deriving the Thevenin and Norton equivalent circuits.

The Norton Equivalent

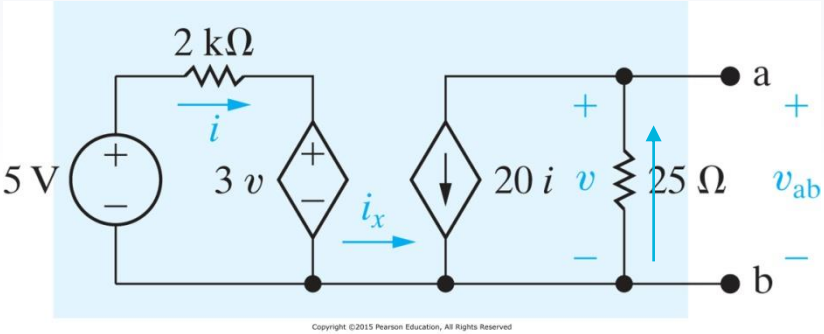
- The Norton equivalent can also easily be derived from the Thevenin equivalent by a source transformation.
- It is also possible to utilize source transformations for deriving the Thevenin and Norton equivalent circuits.
- This approach is useful when the circuit **contains only independent sources**.
- In case the circuit contains **dependents sources** the controlling voltages and/or currents of these sources need to be retained **which may prohibit continued reduction** of the circuit **by source transformations**.



Copyright ©2015 Pearson Education, All Rights Reserved

The Thevenin and the Norton Equivalents

- Example: Find the Thevenin equivalent for the circuit given below.



Copyright ©2015 Pearson Education, All Rights Reserved

- The first step is to **recognize** that the current i_x must be zero.
- The **open circuit or Thevenin voltage** will be the voltage across the 25 Ω resistor.

$$V_{Th} = v_{ab} = (-20i)(25) = -500i$$

- The current i is:

$$i = \frac{5 - 3v}{2000} = \frac{5 - 3V_{Th}}{2000}$$

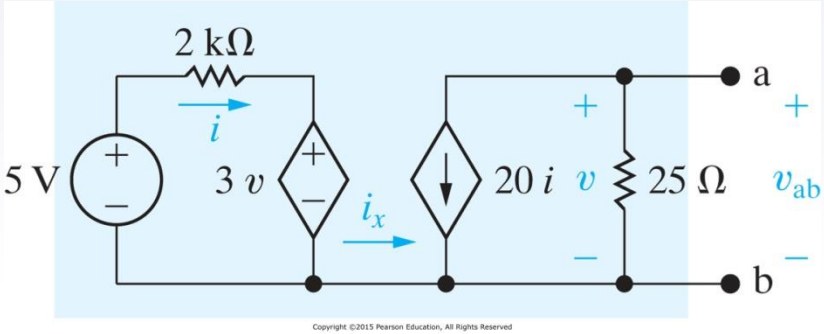
- **Combining** both equations:

$$-\frac{V_{Th}}{500} = \frac{5-3V_{Th}}{2000} \qquad -20V_{Th} = 25 - 15V_{Th} \qquad V_{Th} = -5\ V$$

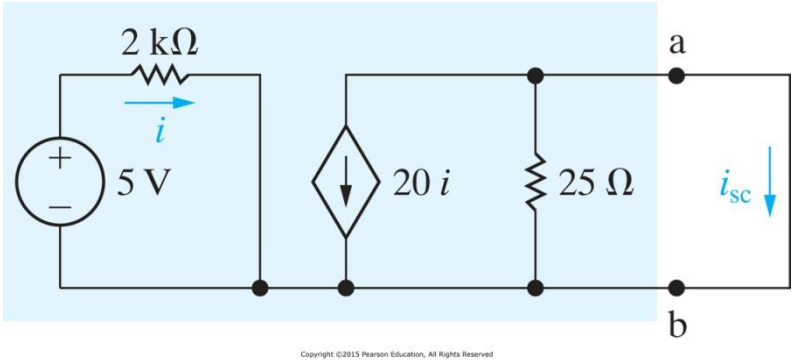
The Thevenin and the Norton Equivalents

10

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved



Copyright ©2015 Pearson Education, All Rights Reserved

- To calculate the short-circuit current, we **place a short circuit** across a, b. When the terminals are shorted together, v is zero.
- In this case, all the current from the dependent source **will flow through the short-circuited path**:

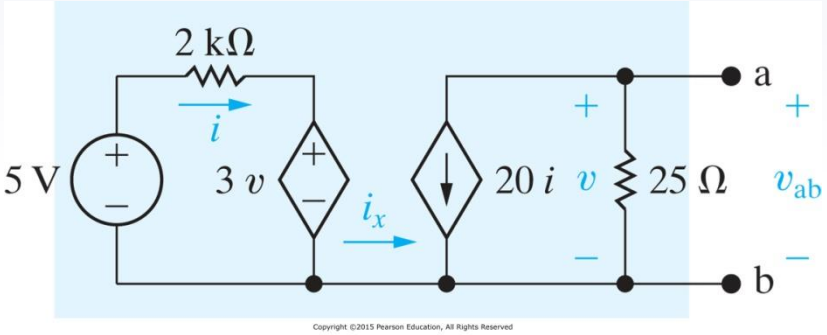
$$i_{sc} = -20i$$

- Because the voltage v, controlling the dependent voltage source has been **reduced to zero**, the current controlling the dependent current source is:

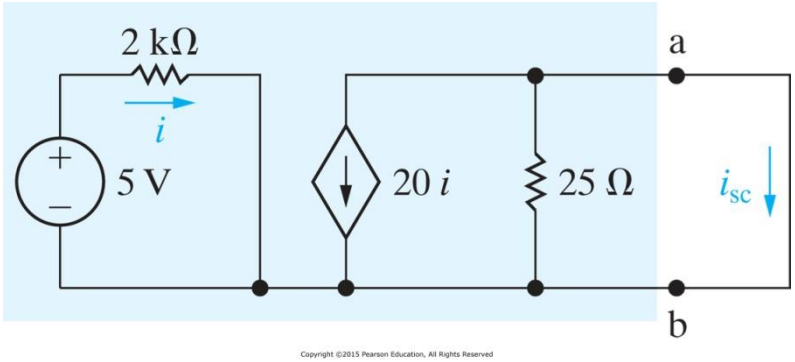
$$i = \frac{5}{2000} = 2.5\ mA$$

The Thevenin and the Norton Equivalents

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved



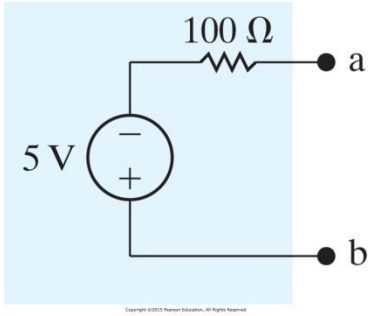
Copyright ©2015 Pearson Education, All Rights Reserved

- **Combining** the two equations,

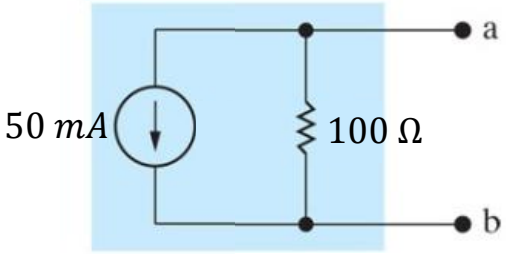
$$i_{sc} = -20(2.5) = -50\ mA$$

- From i_{sc} and V_{Th} we get,

$$R_{Th} = \frac{V_{Th}}{i_{sc}} = \frac{-5}{-50(10^{-3})} = 100\ \Omega$$



The Thevenin equivalent



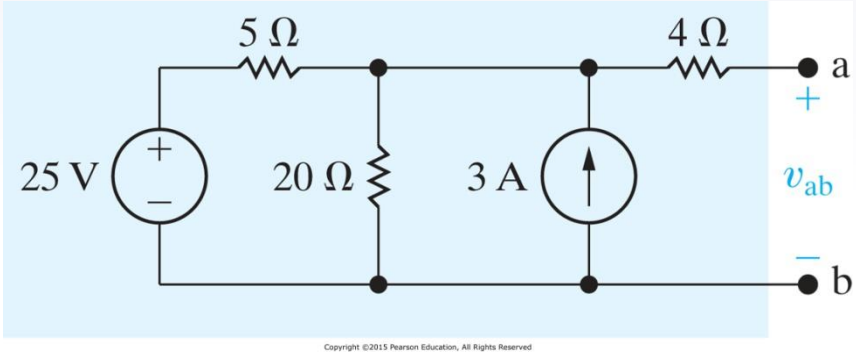
The Norton equivalent

10 The Thevenin and the Norton Equivalents

- It is **also possible** to calculate the **Thevenin resistance** $\boldsymbol{R_{Th}}$ using alternative (**and easier**) calculations.
- If the circuit contains **only independent sources**
  - **Deactivate** all the sources
  - Calculate the equivalent resistance **seen from the given terminals**
- **Deactivating a current source** is equivalent to **replacing** it with **an open circuit**.
- **Deactivating a voltage source** is equivalent to **replacing** it with **a short circuit**.

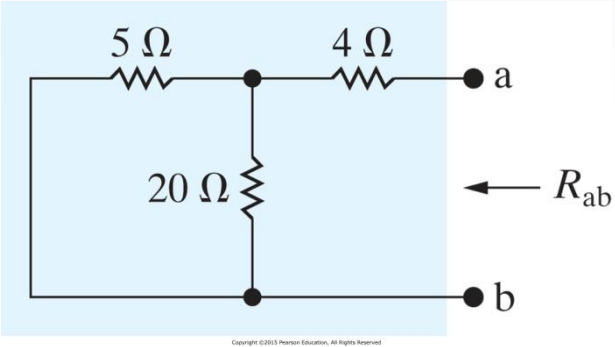
10 The Thevenin and the Norton Equivalents

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved

- **Deactivating** the independent sources, we get:



Copyright ©2015 Pearson Education, All Rights Reserved

- **The resistance seen** looking from the terminals a, b:

$$R_{ab} = R_{Th} = 4 + \frac{5(20)}{25} = 8\ \Omega$$

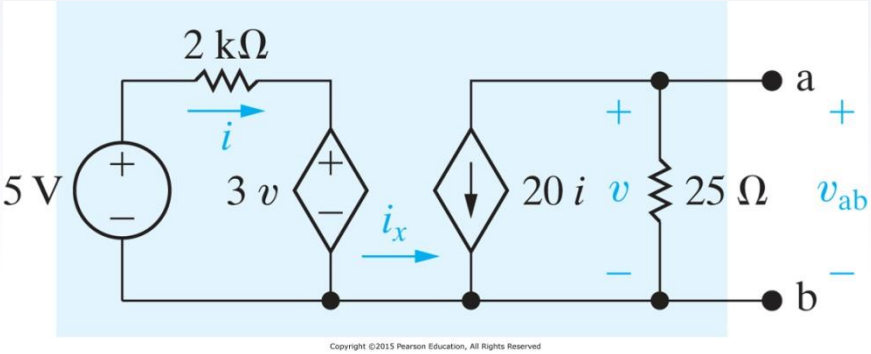
10 The Thevenin and the Norton Equivalents

- In case there are **dependent sources** in the circuit, in order to calculate the Thevenin resistance, one can use the approach,
  - **Deactivate** all the **independent** sources
  - **Apply a test current or voltage source** to the given terminals
  - $R_{Th} = V_{Test} / I_{Test}$ = Voltage across the terminals of the test source / Current through the test source

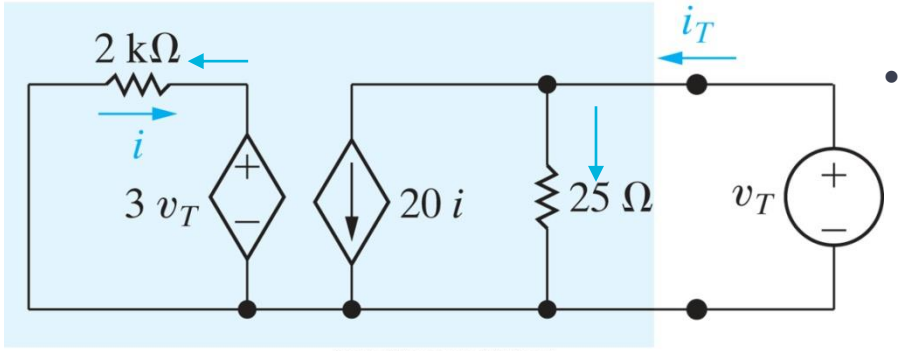
The Thevenin and the Norton Equivalents

10

- Example: Find R_{Th}



Copyright ©2015 Pearson Education, All Rights Reserved



Copyright ©2015 Pearson Education, All Rights Reserved

- We first **deactivate** the independent voltage source from the circuit and then **excite** the circuit from the therminals a, b with either a test voltage source or a test current source.
- If we **apply a test voltage source**, we will **know** the voltage v of the dependent voltage source and we can **easily find** the control current i.
- **Denoting** the current delivered by the test voltage source to the circuit as i_T,

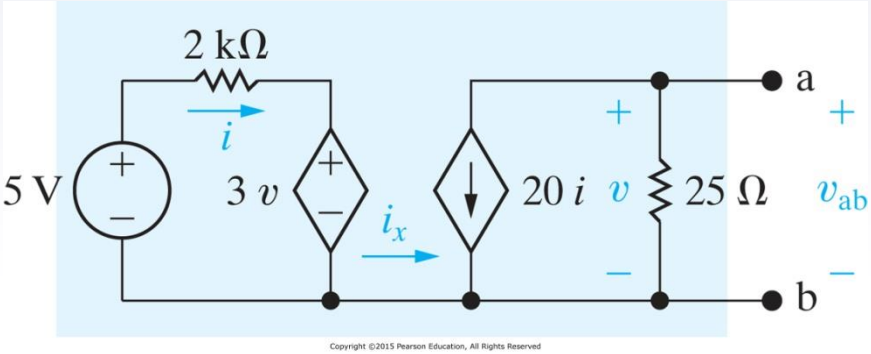
$$i_T = \frac{v_T}{25} + 20i$$

$$i = \frac{-3v_T}{2}\ mA$$

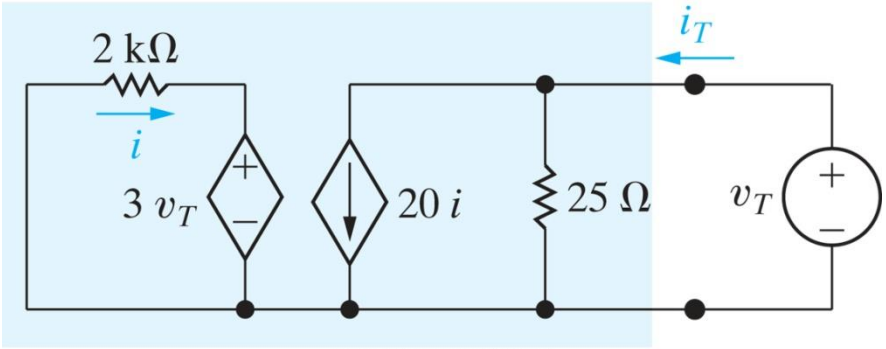
The Thevenin and the Norton Equivalents

10

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved



Copyright ©2015 Pearson Education, All Rights Reserved

- **Substituting** the expression of i into the equation of i_T,

$$i_T = \frac{v_T}{25} - \frac{60v_T}{2000}$$

$$\frac{i_T}{v_T} = \frac{1}{25} - \frac{6}{200} = \frac{50}{5000} = \frac{1}{100}$$

$$R_{Th} = \frac{v_T}{i_T} = 100\ \Omega$$

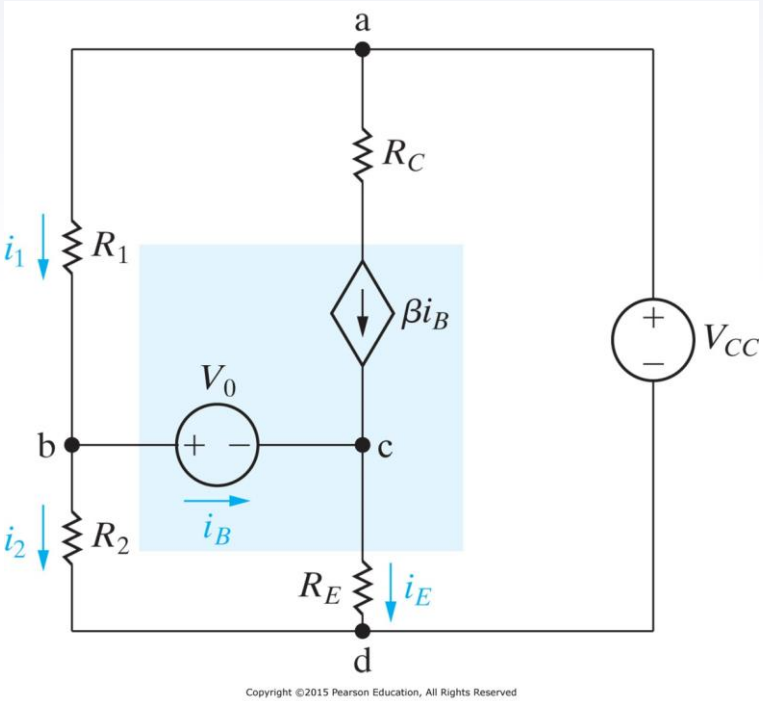
10 The Thevenin and the Norton Equivalents

- In case the circuit **contains resistors and dependent sources only** (and there are **no independent sources**) one needs to **use a test source**.
- Sometimes involving test sources may **result in easier calculations**.
- The values of the introduced test sources (i_T or v_T) can be chosen **arbitrarily**. Usually **a number which will result in easier calculations** are chosen (such as $\boldsymbol{i_T = 1\ A}$ **or** $\boldsymbol{v_T = 1\ V}$).
- The Thevenin and the Norton equivalents are **important tools** which can be used to simplify circuits.
- They can be used **in portions of circuits as well**.

The Thevenin and the Norton Equivalents

10

- Example:



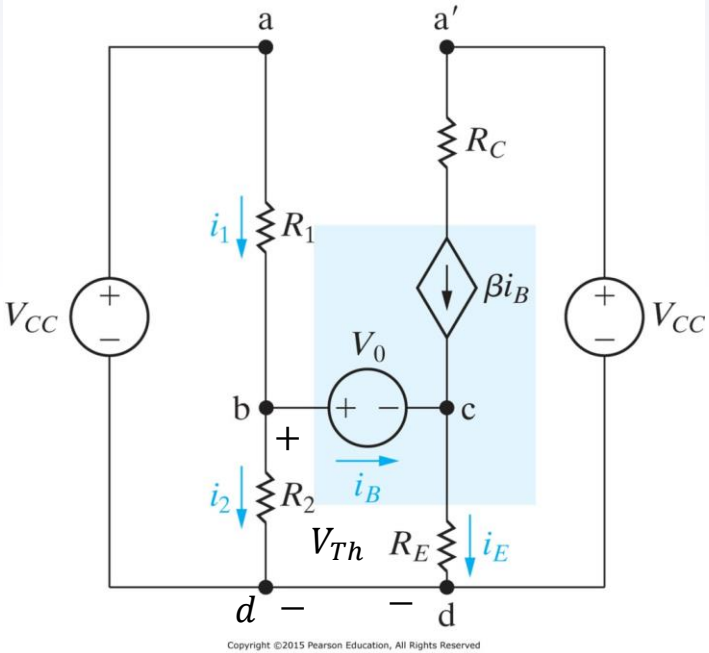
Copyright ©2015 Pearson Education, All Rights Reserved

- In this example circuit i_B is the **key to finding** the other branch currents.

The Thevenin and the Norton Equivalents

10

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved

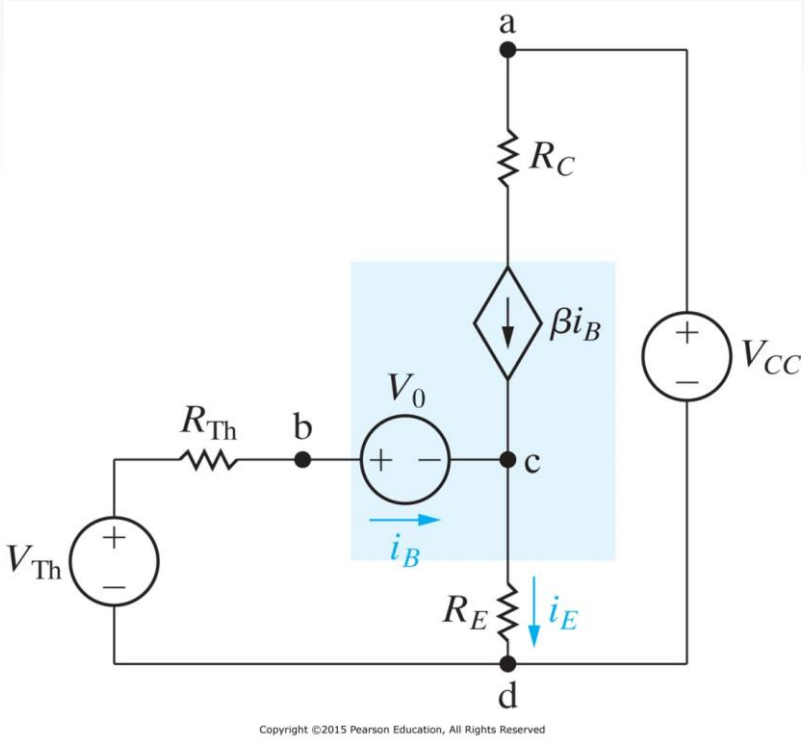
- We first redraw the circuit to prepare to replace the subcircuit to the left of V_o with its Thevenin equivalent.
- This modification has **no effect on the branch currents** i_B, i_1, i_2 and i_E.
- We want to replace the circuit made up of V_{CC}, R_1 and R_2 with a Thevenin equivalent, with respect to the terminals b, d.

$$V_{Th} = \frac{V_{CC}R_2}{R_1 + R_2}$$

$$R_{Th} = \frac{R_1R_2}{R_1 + R_2}$$

The Thevenin and the Norton Equivalents

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved

- We derive an equation for i_B by summing the voltages around the left mesh. While writing this equations, we **recognize** that $i_E = (1 + \beta)i_B$. Thus,

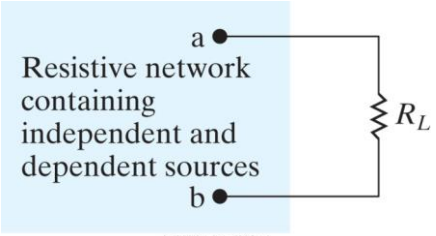
$$V_{Th} = R_{Th}i_B + V_0 + R_E(1 + \beta)i_B$$

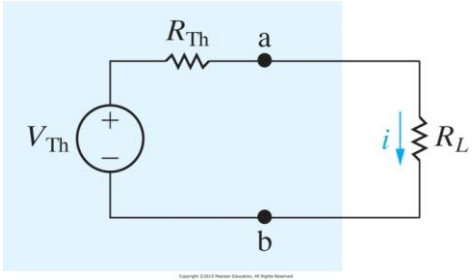
$$i_B = \frac{V_{Th} - V_O}{R_{Th} + (1 + \beta)R_E}$$

- After **incorporating** the Thevenin equivalent into the original circuit, we can **obtain** the solution for i_B by writing a single equation.

Maximum Power Transfer

- Sometimes there is a need to **design** a system **to transfer maximum power** to its load,
  - Power utility systems (**efficiency of power transfer** has particular importance)
  - Communication systems (**the amount of power transferred** has particular importance)
- Here we will analyze the conditions for transferring maximum power from a resistive network/circuit to a load resistance.
- Consider a resistive network connected to a load R_L.
- The aim is to find **the value of R_L that permits maximum power delivery to R_L.**





- The network can be represented by its **Thevenin equivalent**.
- The power dissipated by the load resistor can be easily calculated.

Maximum Power Transfer

11

- **Replacing** the original network by its Thevenin equivalent greatly simplifies the task of finding R_L.
- Derivation of the value of R_L requires expressing **the power dissipated** p as a function of the three circuit parameters V_{Th}, R_{Th} and R_L.

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

- Next, we **recognize** that for a given circuit, V_{Th} and R_{Th} **will be fixed**. Therefore the power dissipated is **a function of the variable R_L**.
- Maximum power transfer **occurs** when **the value of the load resistance R_L is equal to the Thevenin resistance R_{Th}**. In this case, the maximum power delivered to R_L

$$p_{max} = \frac{V_{Th}^2 R_L}{(2R_L)^2} = \frac{V_{Th}^2}{4R_L}$$

Maximum Power Transfer

11

Derivation:

- We use elementary calculus to find the value of R_L that maximizes the power.
- The derivative of p with respect to R_L:

$$\frac{dp}{dR_L} = V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - R_L(2)(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right]$$

- The derivative is zero and p is maximized when

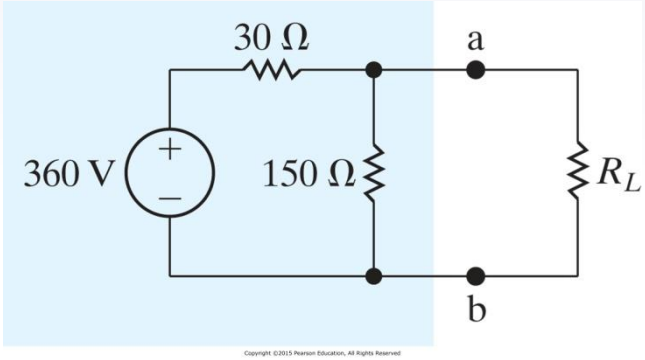
$$(R_{Th} + R_L)^2 = 2R_L(R_{Th} + R_L)$$

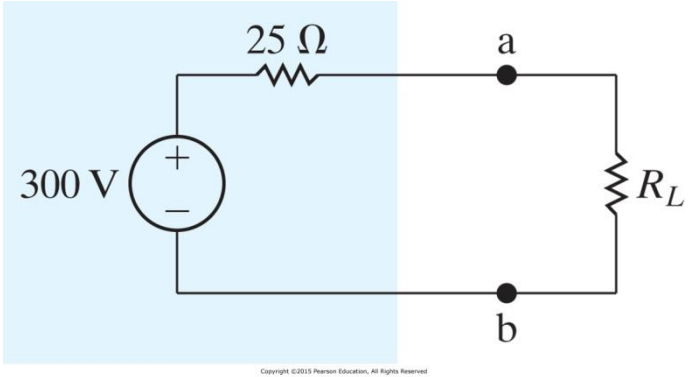
- Solving this equation gives

$$R_L = R_{Th}$$

Maximum Power Transfer

- Example: a) Find the value of R_L that results in maximum power being transferred to R_L.





- The Thevenin voltage for the circuit to the left of the terminals a, b is,

$$V_{Th} = \frac{150}{180}(360) = 300\,V$$

- The Thevenin resistance is,

$$R_{Th} = \frac{(150)(30)}{150 + 30} = 25\ \Omega$$

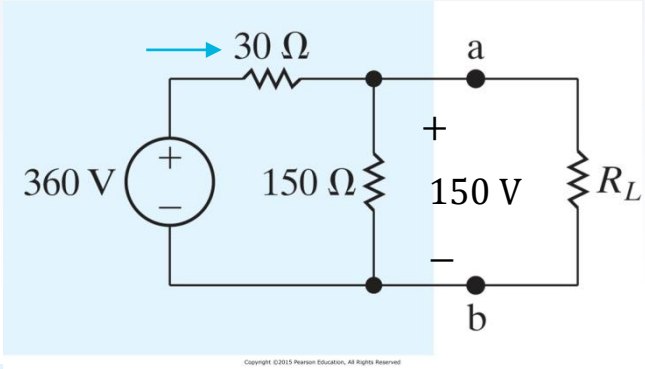
b) Calculate the maximum power that can be delivered to R_L.

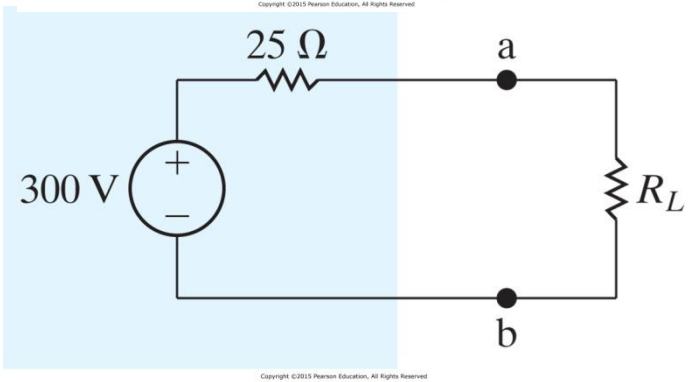
$$p_{max} = \frac{V_{Th}^2 R_L}{(2R_L)^2} = \frac{(300)^2}{(50)^2}(25) = 900\ \text{W}$$

Maximum Power Transfer

11

- Example:





c) When R_L is adjusted for maximum power transfer, what percentage of power delivered by the 360 V source reaches R_L?

- When R_L equals 25 Ω, the voltage, v_{ab} is,

$$v_{ab} = \left(\frac{300}{50}\right)(25) = 150 \text{ V}$$

- The current in the voltage source, in the direction of voltage rise across the source is,

$$i_s = \frac{360 - 150}{30} = \frac{210}{30} = 7\ A$$

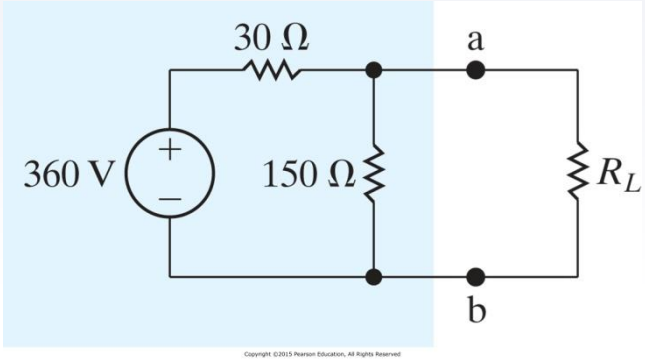
- The power delivered by the source to the circuit,

$$p_s = -i_s(360) = -2520\ W$$

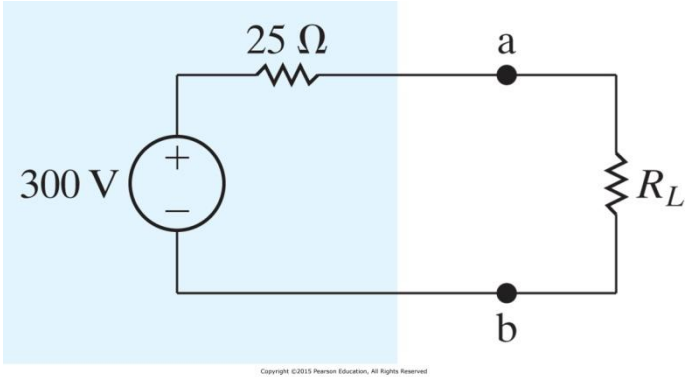
Maximum Power Transfer

11

- Example:



Copyright ©2015 Pearson Education, All Rights Reserved



Copyright ©2015 Pearson Education, All Rights Reserved

$$p_s = -i_s(360) = 2520\ W$$

- The percentage of the source power delivered to the load is,

$$\frac{900}{2520}(100) = 35.71\%$$

EE2001 Electrical Circuits- I:

Continuation of the topic:
The Operational Amplifier

12 December 2023

Outline



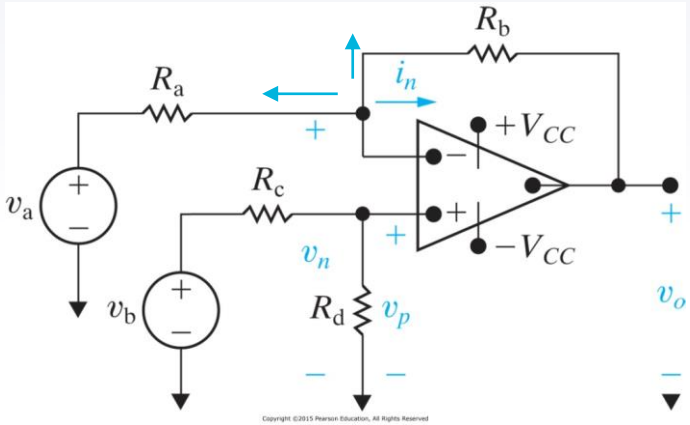
7. The Difference-Amplifier Circuit

8. A More Realistic Model for the Operational Amplifier

The Difference Amplifier Circuit

7

- The difference amplifier takes the difference between two input voltages.
- Consider the following circuit with the op-amp **assumed to be ideal**.



$$i_p = i_n = 0$$

$$v_p = v_n$$

$$v_p = \frac{R_d}{R_c + R_d} v_b$$

$$\frac{v_n - v_a}{R_a} + \frac{v_n - v_o}{R_b} = 0$$

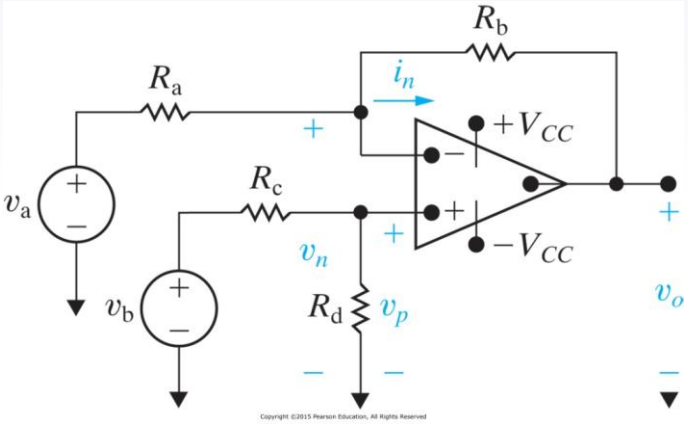
$$v_o = \left(1 + \frac{R_b}{R_a}\right) v_n - \frac{R_b}{R_a} v_a$$

$$v_o = \left(\frac{R_a + R_b}{R_a}\right)\left(\frac{R_d}{R_c + R_d}\right) v_b - \frac{R_b}{R_a} v_a$$

- The **output voltage** is the **<u>weighted difference between the two input voltages.</u>**

The Difference Amplifier Circuit

- The scaling factors (i.e., the weights) of the two voltages in the difference amplifier are in general not equal.
- It is possible to derive a condition for them to be equal.



$$\left(\frac{R_a + R_b}{R_a}\right)\left(\frac{R_d}{R_c + R_d}\right) = \frac{R_b}{R_a}$$

$$\left(1 + \frac{R_b}{R_a}\right) = \frac{R_b}{R_a}\left(1 + \frac{R_c}{R_d}\right)$$

$$1 = \frac{R_b}{R_a}\frac{R_c}{R_d}$$

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

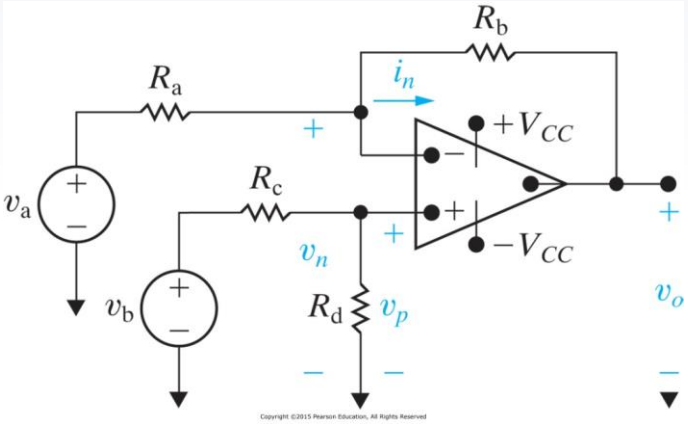
$$v_o = \frac{R_b}{R_a}(v_b - v_a)$$

- If the condition here is satisfied, the output voltage is the scaled difference of the input voltages.
- R_b/R_a is the gain of the difference amplifier.

The Difference Amplifier Circuit

7

- In addition to the condition on the previous page if $R_b = R_a$ in the difference amplifier, then the output voltage is expressed as just the difference between the input voltages.



$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

$$R_b = R_a$$

$$v_o = v_b - v_a$$

- As was in all previous amplifiers the scaling of the input voltages in the difference amplifier are determined by the resistors and can be adjusted by proper choice.

The Common and Difference Modes in the Difference Amplifier

7

- The differential mode of an amplifier is defined as,

$$v_{dm} = v_b - v_a$$

- The common mode of an amplifier is defined as,

$$v_{cm} = \frac{v_a + v_b}{2}$$

- In terms of these variables, one can express v_a and v_b as,

$$v_a = v_{cm} - \frac{1}{2}v_{dm} \qquad v_b = v_{cm} + \frac{1}{2}v_{dm}$$

- Using these the output voltage of the difference amplifier can be expressed as (after some derivations),

$$v_o = \underbrace{\left(\frac{R_aR_d - R_bR_c}{R_a(R_c + R_d)}\right)}_{A_{cm}} v_{cm} + \underbrace{\left(\frac{R_d(R_a + R_b) + R_b(R_c+R_d)}{2R_a(R_c + R_d)}\right)}_{A_{dm}} v_{dm}$$

The Common and Difference Modes in the Difference Amplifier

7

$$v_o = A_{cm}v_{cm} + A_{dm}v_{dm}$$

- A_{cm}= the common mode gain
- A_{dm}= the differential mode gain

$$A_{cm} = \frac{R_aR_d - R_bR_c}{R_a(R_c+R_d)} \qquad A_{dm} = \frac{R_d(R_a+R_b)+R_b(R_c+R_d)}{2R_a(R_c+R_d)}$$

- For $\frac{R_a}{R_b} = \frac{R_c}{R_d}$

  we have

$$A_{cm} = 0 \qquad A_{dm} = \frac{R_b}{R_a}$$

- Then the output voltage is given by

$$v_o = 0.\, v_{cm} + \frac{R_b}{R_a} v_{dm}$$

The Common and Difference Modes in the Difference Amplifier

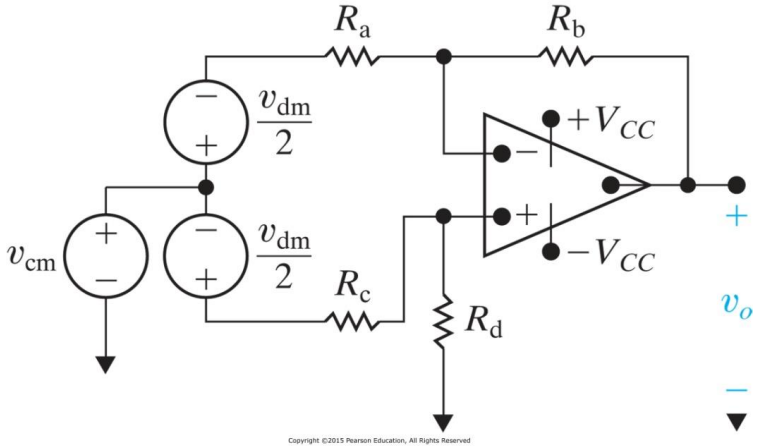
7

- The ideal difference amplifier has $A_{cm} = 0$.

  - It amplifies only the differential mode portion of the input voltage and eliminates the common mode portion of the input voltage.

- In many applications the differential mode portion contains information of interest, whereas the common mode portion contains noise and needs to be eliminated.

- In terms of v_{cm} and v_{dm} the circuit can be redrawn as



Copyright ©2015 Pearson Education, All Rights Reserved

The Common Mode Rejection Ratio

- In practice, the common mode gain A_{cm} of the difference amplifier may not be zero due to
  - Resistance mismatch
  - Non-ideal op-amp
- Lets consider the effect of resistance mismatches.
- Assume that the chosen resistors do not satisfy the below condition exactly.

$$\frac{R_a}{R_b} = \frac{R_c}{R_d}$$

- Instead, assume that the following relation is satisfied (due to resistance mismatch), where ε is a very small constant.

$$\frac{R_a}{R_b} = (1 - \varepsilon)\frac{R_c}{R_d}$$

- Under this condition, the new expressions for the common mode gain A_{cm} and the differential mode gain A_{dm} can be derived as follows.

End of the Lesson (11.12.23)

Any questions?

New supplementary videos are added to the playlist on Youtube:

Supplementary videos: Enter Youtube-> Search for ‘Alparslan Zehir’ -> Go to Playlists -> Watch ‘Electrical Circuits’ playlist